8 Machine Learning: Data Cleaning & Normalization

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What is Data Cleaning?

- Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.
- There are many opportunities for data to be duplicated or mislabeled.
- If data is incorrect, outcomes and algorithms are unreliable, even though they may look correct.
- There is no one absolute way to prescribe the exact steps in the data cleaning process because the processes will vary from dataset to dataset.
- False conclusions because of incorrect or "dirty" data can inform poor business strategy and decision-making.

Cause!!

- Human error during data entry, recording, or encoding
- A faulty sensory device as is the case in the IoT
- Corruption is a result of hardware or software damage.

How to clean data

- While the techniques used for data cleaning may vary according to the types of data, you can follow these basic steps to map out a framework for data cleaning.
  1. Remove duplicate or irrelevant observations
  2. Fix structural errors
  3. Filter unwanted outliers
  4. Handle missing data
  5. Validate and Q&A

1. Remove duplicate or irrelevant observations

- Remove unwanted observations from your dataset, including duplicate observations or irrelevant observations.
  - Collect from many places may be redundant
  - Some data do not fit into the specific problem
  - Older generations/ Obsolete data
- This can make analysis more efficient and minimize distraction from the primary target—as well as create a more manageable and more performant dataset.

2. Fix structural errors

- Structural errors are when you measure or transfer data
- Notice strange naming conventions, typos, or incorrect capitalization.
- These inconsistencies can cause mislabeled categories or classes.
  - For example, you may find "N/A" and "Not Applicable" both appear, but they should be analyzed as the same category.

3. Filter unwanted outliers

- Often, there will be one-off observations where, at a glance, they do not appear to fit within the data you are analyzing.
- If you have a legitimate reason to remove an outlier, like improper data entry, doing so will help the performance of the data you are working with.
- Remember: just because an outlier exists, doesn't mean it is incorrect. This step is needed to determine the validity of that number. If an outlier proves to be irrelevant for analysis or is a mistake, consider removing it.

4. Handle missing data

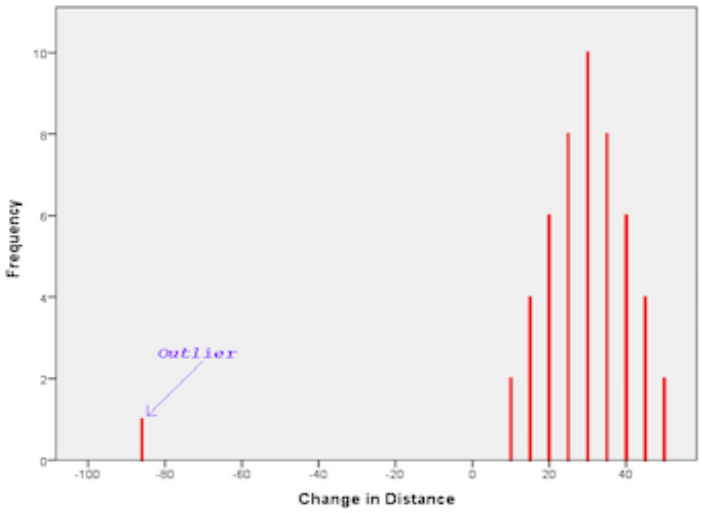
- You can't ignore missing data because many algorithms will not accept missing values. There are a couple of ways to deal with missing data. Neither is optimal, but both can be considered.
  - As a first option, you can drop observations that have missing values, but doing this will drop or lose information, so be mindful of this before you remove it.
  - As a second option, you can input missing values based on other observations; again, there is an opportunity to lose the integrity of the data because you may be operating from assumptions and not actual observations.
  - As a third option, you might alter the way the data is used to effectively navigate null values.

5. Validate and Q&A

- At the end of the data cleaning process, you should be able to answer these questions as a part of basic validation:
  - Does the data make sense?
  - Does the data follow the appropriate rules for its field?
  - Does it prove or disprove your working theory, or bring any insight to light?
  - Can you find trends in the data to help you form your next theory?
  - If not, is that because of a data quality issue?

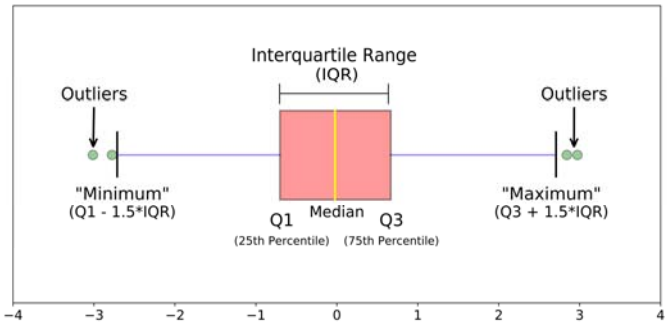
What are outliers?

- Outliers are extreme data points. They are usually very low or very high values that do not conform to the rest of the pattern in the data, for example:
  - A customer who is 200 years of age
  - A sensor reading of 40° Celsius in the Antarctica region

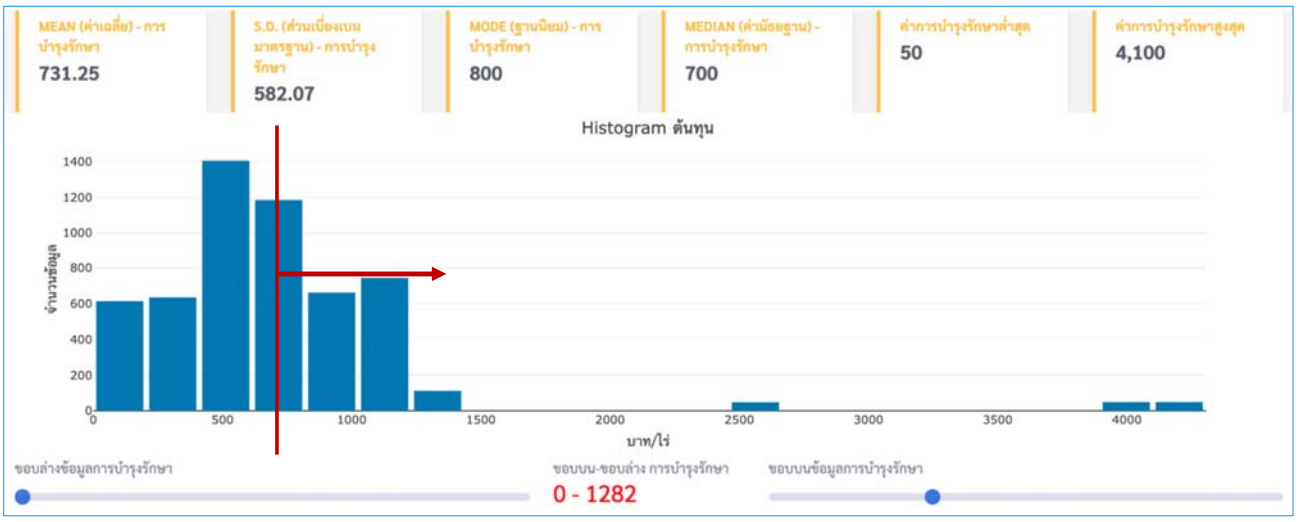


Handling Outliers

- **Identifying Outliers** There are different techniques to identify outliers either through visual means or statistical techniques
  - Visualization: such as boxplots, histograms and scatterplots can make it easy to detect outliers.
  - Using Z-scores: observe data follows a normal distribution, standard deviations by which observed values fall above or below a mean value
  - Using Quartile, Decile or Percentile as shown in figure



Example

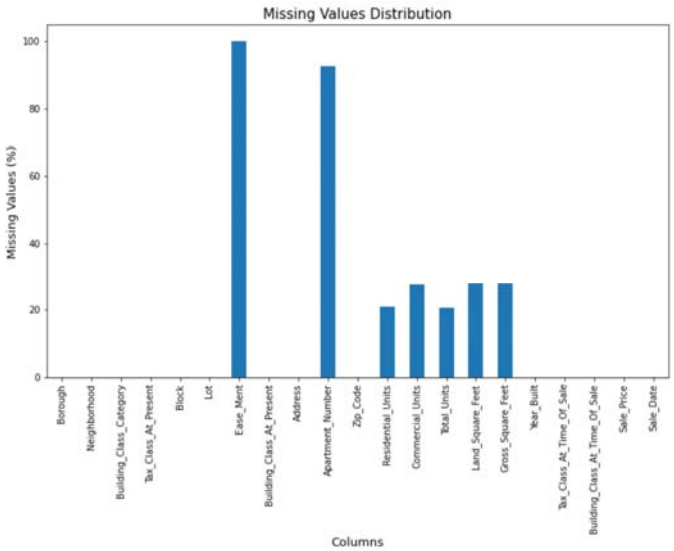


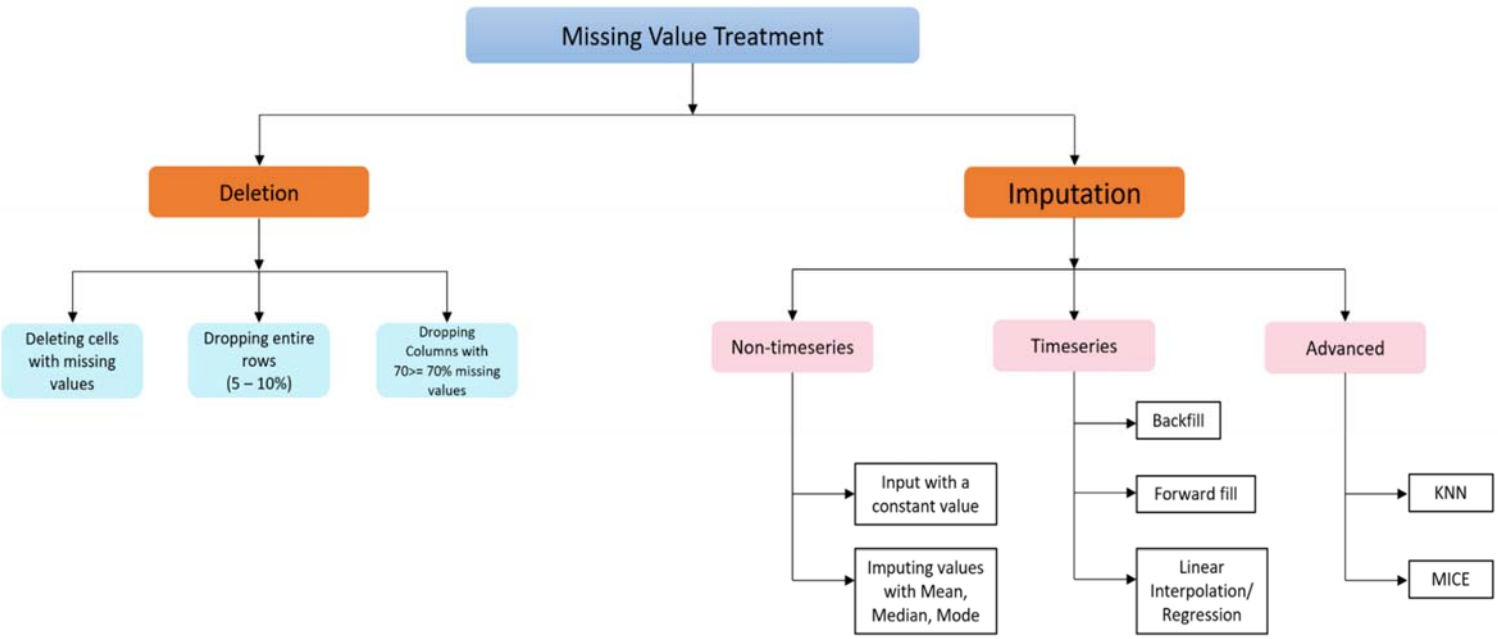
- http://www.icicel.org/ell/contents/2023/1/el-17-01-01.pdf

What are missing values?

- There are times when there are observations (rows) with missing data and sometimes you can have entire columns with completely missing values.
- So what do we do in instances like this? Do we ignore it? This can be likened to a leaking roof and when it rains, it leaks.
- Missing values have different representations across different tools. These representations include NULL, empty strings, NaN, NA, #NA, etc.
- Data context is important in dealing with missing data, and understanding why data is missing is crucial.
- There are two primary methods for dealing with missing data:
  - Removal of data
  - Imputation of data

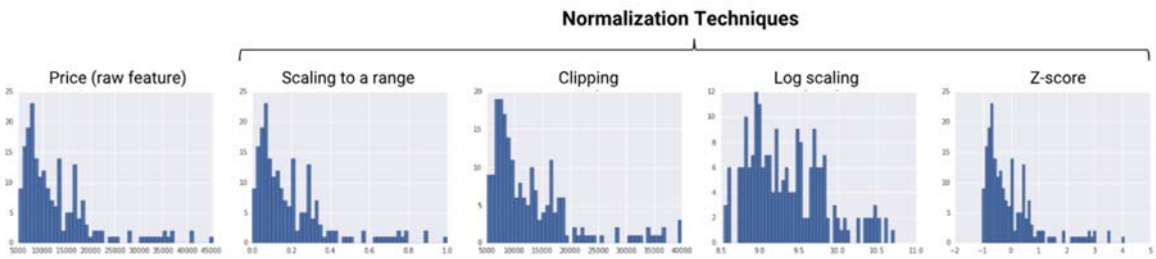
- Before any decision can be made on what method will be used, one must first examine the patterns of missing values across columns. We will use visual and descriptive techniques to assess these





Normalization

- Data normalization is a crucial element of data analysis.
- It's what allows analysts to compile and compare numbers of different sizes, from various data sources.
- Common normalization techniques:
  - Scaling to a range, Linear normalization (or "Max-Min,")
  - Clipping
  - Log scaling
  - Z-Score



https://developers.google.com/machine-learning/data-prep/transform/normalization

Max-Min Normalization

- Formula:

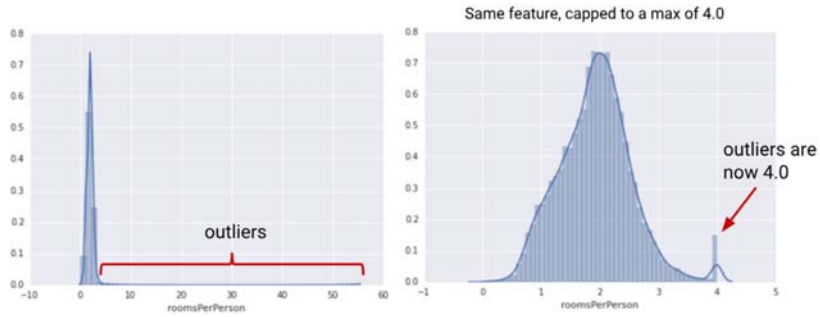
$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

- The scale of this value is between 0-1
- If you want to normalize to 100, we just have to multiply the fraction by the number needed to get the denominator to 100.
- Example: $x_{min} = 5, x_{\max} = 50, x = 20$

$$x' = \frac{20 - 5}{50 - 5} = 0.3$$

Clipping

- If the data set contains extreme outliers, you might try feature clipping, which caps all feature values above or below a certain value to a fixed value.
- For example: clip all temperature values above 40 to be exactly 40.
- Apply feature clipping before or after other normalizations.
- Formula: Set min/max values to avoid outliers.
- Another simple clipping strategy is to clip by z-score to $\pm N\sigma$ (e.g. limit to $\pm 3\sigma$). Note that σ is the standard deviation.

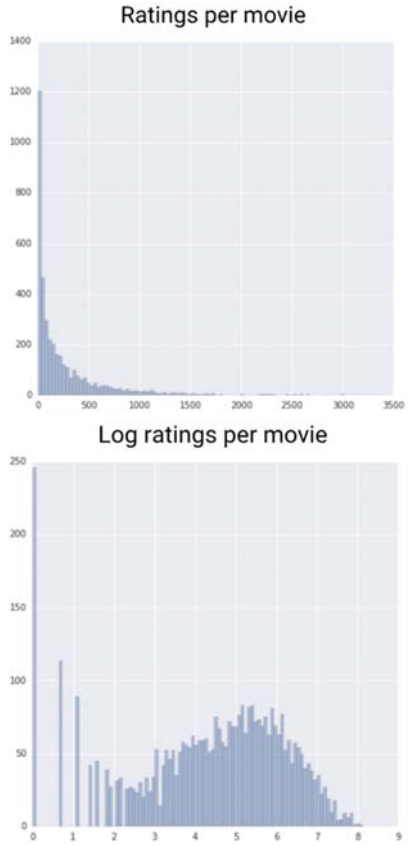


Log Scaling

- Log scaling computes the log of your values to compress a wide range to a narrow range.

$$x' = \log x$$

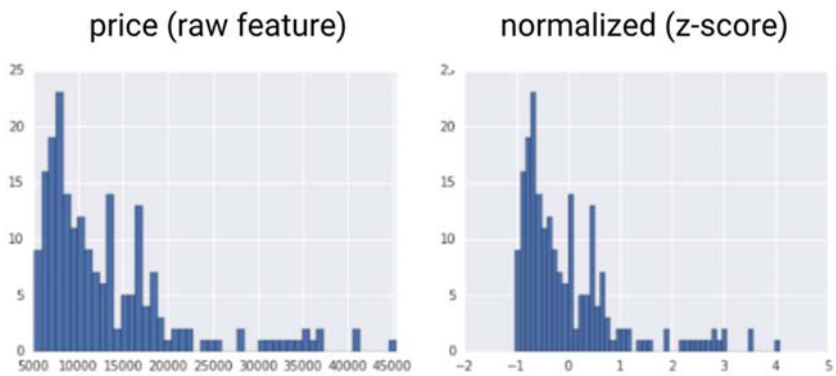
- Log scaling is helpful when a handful of your values have many points, while most other values have few points. This data distribution is known as the *power law* distribution.
- Example: Movie ratings, most movies have very few ratings (in the tail), while a few have lots of ratings (in the head). Log scaling changes the distribution, helping to improve linear model performance.



Z-Score

- Z-score is a variation of scaling that represents the number of standard deviations away from the mean (mean = 0 and std = 1).
- It's useful when there are a few outliers, but not so extreme that you need clipping.
- The formula for calculating the z-score of a point, x, is as follows:

$$x' = \frac{x - \mu}{\sigma}$$



Summary

Normalization Technique	Formula	When to Use
Linear Scaling	$x' = (x - x_{min})/(x_{max} - x_{min})$	When the feature is more-or-less uniformly distributed across a fixed range.
Clipping	if x > max, then x' = max. if x < min, then x' = min	When the feature contains some extreme outliers.
Log Scaling	x' = log(x)	When the feature conforms to the power law.
Z-score	x' = (x - μ) / σ	When the feature distribution does not contain extreme outliers.